



# Exploring Data Transformation with Google Cloud

This study guide covers the materials from the second course of the six-course series for the Cloud Digital Leader certification.

## Certification exam focus

The Cloud Digital Leader certification exam is designed to assess knowledge of cloud literacy, common business use cases, and how cloud solutions and the capabilities of Google Cloud products support an enterprise.

### The value of data

Data represents raw information—both structured and unstructured—that an organization collects from every interaction and process. The importance of data lies in its ability to empower smarter, real-time decision-making and personalized customer experiences.

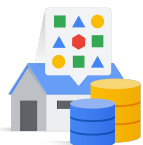


By leveraging an intelligent data cloud to analyze everything from financial reports to social media sentiment, organizations can perceive emerging trends, predict future needs, and maintain a competitive edge in a rapidly changing market.

### Data storage



**Database:** An organized collection of data stored in tables and accessed electronically from a computer system.



**Data warehouse:** An enterprise system used for the analysis and reporting of structured and semistructured data from multiple sources.

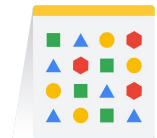
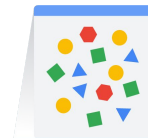
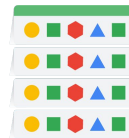


**Data lake:** A repository designed to ingest, store, explore, process, and analyze any type or volume of raw data regardless of the source.

It is important to maintain openness and interoperability with data management platforms. This allows data to move freely between different tools, ensuring it doesn't get trapped in isolated systems or tied to one specific provider.

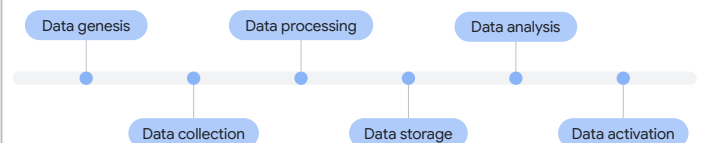
### Types of data

- **First-party:** Proprietary customer datasets collected directly by a business through its own audience transactions and digital interactions.
- **Second-party:** Data owned by another organization, such as a partner, that is used to augment a company's own internal datasets.
- **Third-party:** External datasets collected by organizations with no direct customer relationship; often purchased to provide broader context and insights.



- **Structured:** Highly organized and well-defined information typically stored in tables with clear relationships between rows and columns.
- **Unstructured:** Information that lacks a predefined data model or organization, including formats like text and video files, which historically required machine learning to analyze.
- **Semistructured:** Data that falls between categories, lacking a formal database structure but containing tags such as in JSON files, making it easier to analyze than unstructured data.

### Data supply chain



The steps of the data supply chain include data genesis, data collection, data processing, data storage, data analysis, and data activation.





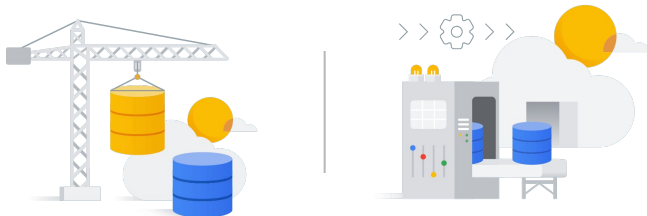
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## Key data management terms

- **Relational:** Organizes structured data into rows and columns within tables. It uses a predefined schema, defines relationships via keys, and is managed using SQL.
- **Non-relational:** Database approach designed to handle large volumes of rapidly changing data that doesn't fit well into the rigid, tabular structure of a relational database.
- **Object storage:** Data storage architecture that manages data as objects in a flat address space rather than in a file hierarchy. This is ideal for unstructured data at massive scale.
- **SQL:** The standard, declarative language used to manage, query, and manipulate relational databases.
- **NoSQL:** A term for database systems that do not use SQL as their primary query language. NoSQL offers flexible schema models, often prioritizing availability and scalability over strict consistency.

## Database migration and modernization

Organizations can modernize databases via a simple lift-and-shift for minimal upheaval or choose a fully managed migration using tools like Google Cloud's Database Migration Service and Datastream to offload maintenance.



## Google Cloud data management



### Cloud Storage

Object storage service for unstructured data, like images, videos, logs, and backups.



### Cloud Spanner

Relational database service that offers both traditional relational structure and horizontal scalability.



### Cloud SQL

Fully managed, relational database service for MySQL, PostgreSQL, and SQL Server databases on Google Cloud.



### AlloyDB

Fully managed, PostgreSQL-compatible database service for demanding enterprise workloads.



### Bigtable

Fully managed, NoSQL database service for large analytical and operational workloads.



### BigQuery

Serverless, highly scalable, and cost-effective cloud data warehouse.



### Firestore

Flexible, scalable, NoSQL document database for mobile, web, and server development.

## Cloud Storage classes

- **Standard:** Highest storage cost but no retrieval fees. Best for data accessed frequently.
- **Nearline:** Lower storage cost with a small retrieval fee. Ideal for data accessed about once a month.
- **Coldline:** Even lower storage cost but higher retrieval fees. Best for data accessed about once a quarter.
- **Archive:** Lowest storage cost with the highest retrieval fees. Best for data accessed less than once a year.
- **Autoclass:** Automatically transitions objects between classes based on access patterns to optimize costs.





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## Looker

Looker is a web-based Google Cloud business intelligence platform that democratizes data by allowing anyone to explore data and answer their own questions without technical expertise, empowering organizations to uncover and act on insights immediately.

Pairing Looker with SQL databases, like Google Cloud's BigQuery, allows organizations to create real-time reports and dashboards and integrate data into workflows.



## Streaming analytics

Streaming analytics enables businesses to process and analyze data records continuously as they are generated rather than waiting for traditional delayed batches. This real-time approach allows organizations to react instantly to time-sensitive events, ensuring decisions are made using the freshest data possible.

### Use cases include:



**Ecommerce:** User clickstreams can be analyzed to optimize the shopping experience with real-time pricing, promotions, and inventory management.



**Financial services:** Account activity can be analyzed to detect abnormal behaviors in the data stream and generate a security alert.



**News media:** User click records can be streamed from various news source platforms, and data can then be enriched with demographic information to better serve articles that are relevant to the targeted audience.

## Data governance



Data governance is a framework of processes, roles, and standards that ensures information is used effectively to achieve organizational goals. It establishes clear rules for who can access and manage data, as well as how and when they can do so.

## Google Cloud's data processing tools



Pub/Sub

A fully managed, asynchronous messaging service used to publish and subscribe to real-time data streams. It ensures reliable event delivery and seamless data distribution for streaming analytics.



Dataflow

A fully managed service for executing data processing pipelines. It automatically scales resources, optimizes processing, and is built on the open-source Apache Beam model.



Managed Service for  
Apache Spark

A fully managed Apache Spark and Apache Hadoop service on Google Cloud. It allows users to run large-scale data processing jobs with reduced overhead for cluster management.

